Deep Equivariant Multi-Agent Control Barrier Functions

A Study Report- Prajith S

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Overview

The paper introduces *symmetry-infused multi-agent control barrier functions* (CBFs) that guarantee forward invariance of a safe set, argues why exploiting symmetries improves data-driven distributed CBFs and policy generalisation, and gives a recipe for graph networks that respect the system's symmetries.

Conventions. Numbering follows the paper. Boxes: **Intuition** (worked example), **Note** (caveat or clarification), **Key idea** (takeaway), **Q&A** (a question raised while reading, with its answer). Anything marked "not in the paper" is my own derivation or reading.

1 Preliminaries

1.1 Groups, actions and equivariance

A group $(G, \cdot)$ has an identity e, associative operation and inverses.

Definition 1 (Group action)**.** A family of maps $\phi_g : X \to X$ is an *action* of G on X if $\phi_e(x) = x$ and $\phi_g \circ \phi_h = \phi_{g \cdot h}$.

Example: $G = \{0°, 90°, 180°, 270°\}$ acting on $\mathbb{R}^2$ by rotation; $\phi_{180°} \circ \phi_{90°} = \phi_{270°}$.

Orbit: $\mathcal{O}_x = \{\phi_g(x) \mid g \in G\}$; an action splits X into orbits. **Free:** $\phi_g(x) = x \iff g = e$ (no non-identity element fixes any point). $SO(3)$ on the sphere is *not* free (rotation about the vertical axis fixes the pole). **Transitive:** $\forall x, y\ \exists g : \phi_g(x) = y$ (one orbit). $SO(3)$ on the sphere is transitive.

Definition 2 (Equivariant function)**.** For actions ϕ_g on X and ψ_g on Y, $f : X \to Y$ is *equivariant* iff $\psi_g(f(x)) = f(\phi_g(x))$ for all x, g. E.g. $f(x, y) = (2x, 2y)$ is rotation-equivariant.

Invariant: the output is unchanged by a transformation of the input.
Equivariant: the output transforms correspondingly.

1.2 Manifolds and Lie groups

A *smooth manifold* $\mathcal{X}$ looks locally Euclidean with smoothly compatible charts, so calculus applies. The *tangent space* $T_x\mathcal{X}$ is the set of instantaneous directions of motion at x; a *smooth vector field* assigns to each x a vector in $T_x\mathcal{X}$, and the space of these is written $\mathfrak{X}(\mathcal{X})$. The dimension of a manifold is that of the tangent space at one point.

A **Lie group** is a group that is also a smooth manifold, with smooth multiplication and inversion. Its dimension is that of the underlying manifold (e.g. $\dim SO(3) = 3$: at any orientation there are three independent infinitesimal rotations).

1.3 Homogeneous spaces and torsors

A **homogeneous space** is a manifold admitting a transitive G-action: any point can be mapped to any other, so G is its *symmetry*. Examples: S^2 under $SO(3)$; $\mathbb{R}^2$ under $(\mathbb{R}^2, +)$.

A **torsor** $\mathfrak{G}$ of a Lie group G is its underlying manifold with the group structure "forgotten", on which G acts *freely and transitively*: any point reaches any other by exactly one group element, and there is no distinguished origin. Identifying $\chi_g \in \mathfrak{G}$ with $g \in G$, the action is

$$\boxed{\phi(\hat{g}, \chi_g) \simeq \hat{g}\, g.}$$

Free: $\hat{g}g = g \Rightarrow \hat{g} = e$. *Transitive and unique:* $\hat{g}g = h \iff \hat{g} = hg^{-1}$. (For a transitive action, uniqueness of the mover is equivalent to freeness.) Example: for $(\mathbb{R}, +)$, the point 3 acted on by 5 gives 8.

Note
"A manifold serving as a torsor for multiple Lie groups may admit multiple symmetries": the manifold alone does not fix its symmetry group.

2 Problem Formulation

2.1 Setup

N homogeneous robots, $i \in I_N = \{1, \dots, N\}$, with control-affine dynamics $\dot{x}_i = f(x_i, u_i)$, state $x_i \in \mathcal{X}$ (smooth manifold), input $u_i \in \mathcal{U}$, $f : \mathcal{X} \times \mathcal{U} \to \mathfrak{X}(\mathcal{X})$ locally Lipschitz. A Lie group G acts transitively on $\mathcal{X}$ via ϕ. The state splits as $x_i = [p_i, q_i]$ (position $p_i \in \mathcal{P} \subset \mathcal{X}$, other states q_i); targets are $\hat{x}_i \in \mathcal{X}$.

Each robot has w LiDAR rays $y_j^i \in \mathcal{P}$ with range R, padded as $\bar{y}_j^i = [y_j^i, q_0]$, where $q_0 \in \mathcal{X} \setminus \mathcal{P}$ is a fixed neutral state (e.g. zero velocity). Robots communicate within $\hat{R} > R$:

$$\mathcal{N}_i^{\hat{R}} := \{j \in I_N \setminus \{i\} : \|p_i - p_j\| \le \hat{R}\}.$$

2.2 Graph representation

$\mathcal{G}_t$ has vertices $\{x_i(t)\}$ and edges (i, j), $j \in \mathcal{N}_i^{\hat{R}}$. The **extended graph** $\hat{\mathcal{G}}_t$ adds target nodes $\hat{x}_i$ and obstacle nodes $\bar{y}_j^i(t)$ ($2N + w$ nodes in total):

$$V_{\hat{\mathcal{G}}_t} = V_{\mathcal{G}_t} \cup \{\bar{y}_j^i\} \cup \{\hat{x}_i\}, \qquad E_{\hat{\mathcal{G}}_t} = E_{\mathcal{G}_t} \cup \{(i, j) \mid j \in W\} \cup E_*, \quad E_* = \{(i, l) \mid l \in I_{N+N}\}.$$

Indices: 1..N robots (I_N), N+1..2N targets (I_{N+N}), 2N+1..2N+w LiDAR points (W). (The paper writes $W = \{2N, \dots, 2N + w\}$, which has $w + 1$ elements; I use 2N+1.)

Note
(i) E_* (agent–target) lets the GNN route goal information to π_θ. (ii) Safety is evaluated over $E_{\hat{\mathcal{G}}_t} \setminus E_*$: targets are excluded from collision checks, otherwise a robot could never reach within r of its own goal.

Can obstacles move?
Structurally yes, but the certificate assumes static ones: obstacle nodes are re-measured each step, yet padded with q_0 (no velocity), the CBF sum in Eq. (2) runs over agents only, and experiments use static obstacles. Moving obstacles would need velocity features and an extra term in $\dot{h}$.

2.3 Pose and non-pose variables

Pose $\rho_i = (p_i, R_i)$; **non-pose** variables (twist $\xi_i = (\dot{p}_i, \omega_i)$, etc.). Assume the pose submanifold $\bar{\mathcal{X}} \subseteq \mathcal{X}$ is the torsor $\mathfrak{G}$ of G, so each pose is identified with a group element $g_i \in G$. Non-pose variables v_i are not group elements but *transform compatibly* under G: $\phi_e(v) = v$, $\phi_g \phi_h = \phi_{gh}$, and physically a 90° rotation of the pose must rotate $\dot{p}_i$ by 90° too.

Why a torsor for positions but not velocities?

Positions have no preferred origin; velocities do ($\dot{p} = 0$ means standing still). So translating the origin leaves velocity unchanged.

One group acts on different spaces through different representations:

- **Pose** (p, R), $g = (R_0, p_0)$: $(R_0 p + p_0,\ R_0 R)$.
- **Linear velocity**: $\dot{p} \mapsto R_0 \dot{p}$ (translation acts trivially; follows from $\frac{\mathrm{d}}{\mathrm{d}t}(R_0 p) = R_0 \dot{p}$).
- **Scalars** (mass, distances, node-type tags): unchanged.

Node attributes become $(g_i(t), v_i(t))$: a local frame in G plus an invariant/local feature payload, with a type encoding (robot $[p_i, q_i]$, target $[\hat{p}_i, q_0]$, obstacle $[y, q_0]$). The **agent-i subgraph** $\hat{\mathcal{G}}_t^i$ consists of i, its neighbours $\mathcal{N}_i^{\hat{R}}$, and its target and LiDAR nodes (unlike $\mathcal{N}_i^{\hat{R}}$, it contains i).

2.4 Safety-constrained distributed control

$\bigoplus$ stacks states of a robot subset $\bar{I}_N$: $x^{\bar{I}_N} = \bigoplus_{i \in \bar{I}_N} x_i$, and likewise $u^{\bar{I}_N}$, $f^{\bar{I}_N}$. The paper uses two instances: $\bar{I}_N = I_N$ (global state, safe set) and $\bar{I}_N = \mathcal{N}_i^{\hat{R}} \cup \{i\}$ (local input $x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}}$ of h and π). By Assumption 2, only $\mathcal{N}_i^{R} \cup \{i\}$ actually matters.

Safe set. $S_N^r := \{x^{I_N} \mid \|p_i - p_j\| > r,\ \forall (i,j) \in E_{\hat{\mathcal{G}}_t} \setminus E_*,\ i \neq j\}$.

Why $r \in [0, R)$, and why R rather than $\hat{R}$?

(Not spelled out in the paper.) r is the minimum allowed separation, so S_N^r is a family indexed by r. R is the LiDAR range: if $r = R$ every detectable obstacle would already be a violation, leaving no time to react; $r < R$ leaves a margin $[r, R)$ (cf. Assumption 1). Since $\hat{R} > R$, $r < R$ implies every pair that could violate safety is connected by an edge.

Goal. Design a distributed policy $u_i = \pi(\hat{\mathcal{G}}_t^i)$ such that (1) *liveness:* $\|p_i(t) - \hat{p}_i\| \to 0$, and (2) *safety:* $x^{I_N}(t) \in S_N^r$.

Problem 1. Given N robots and compatible targets, solve

$$u_{0:T}^{I_N,*} = \arg\min_{u_{0:T}^{I_N}} \int_0^T \mathcal{T}_N\big(x^{I_N}(\tau)\big)\mathrm{d}\tau \quad \text{s.t.} \quad \dot{x}_i = f(x_i, u_i),\ u_i = \pi(\hat{\mathcal{G}}_t^i) \in \mathcal{U},\ x^{I_N}(t) \in S_N^r, \tag{1}$$

with $\mathcal{T}_N = \sum_i \|p_i - \hat{p}_i\|$.

It is hard because S_N is not available in closed form and must be decomposed into agent-centric sets, and the policy may only use local information.

Does $\arg\min$ minimise distance to target at every instant?

No: it minimises the *integral* of the distance, one trajectory-level cost (area under the distance curve), not a terminal cost. Being far from the goal at any time is penalised, so the optimum approaches the goal as fast as dynamics and safety allow and then stays. Liveness corresponds to $T \to \infty$. In practice the paper imitates a nominal controller and the QP of Section 4.2 instead.

Assumption 1. R is large enough that Problem 1 stays feasible once an obstacle is detected (the agent has time to stop safely).

3 Safety Certificates for Multi-Agent Systems

CBFs certify forward invariance of a safe set [24].

Definition 3 (Safety of the swarm)**.** The system is safe at t ($x^{I_N}(t) \in S_N^r$) **iff** every local graph is safe: $\forall i$, $x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}}(t) \in S_{N,i}^r := \{x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}} \mid \|p_i - p_j\| > r,\ \forall (i,j) \in E_{\hat{\mathcal{G}}_t^i} \setminus E_*\}$.

Since the local edge sets cover all edges, $\bigcap_i S_{N,i}^r = S_N^r$ once each local set is lifted to the full state space; Definition 4 writes only "$\subseteq$", which is all it needs.

3.1 Distributed CBFs

Assumption 2. The safety of node i is affected only by nodes in $\mathcal{N}_i^R \subseteq \mathcal{N}_i^{\hat{R}}$.

Definition 4 (Distributed CBF)**.** Let $h : \mathcal{X}^{|\mathcal{N}|+1} \to \mathbb{R}$ be C^1, Assumption 2 hold, and $\mathcal{C}_N \subset S_N^r$. Then h is a *valid distributed CBF* if for some locally Lipschitz extended class-$\mathcal{K}$ function α,

$$\sup_{u^{\mathcal{N}_i^{\hat{R}}\cup\{i\}}} \left[\sum_{j\in\mathcal{N}_i^{\hat{R}}\cup\{i\}} \left\langle \nabla_{x_j} h(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}}), f(x_j,u_j)\right\rangle \right] \;\geq\; -\alpha\big(h(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}})\big), \tag{2}$$

where $\mathcal{B}_i := \{x^{\mathcal{N}_i\cup\{i\}} \mid h \geq 0\}$ and $\mathcal{C}_N = \bigcap_{i\in I_N}\{x^{I_N} \mid x^{\mathcal{N}_i\cup\{i\}} \in \mathcal{B}_i\} \subset \bigcap_i S_{N,i}^r \subseteq S_N^r$.

Reading Eq. (2)

The sum is $\dot{h} = \sum_j \langle \nabla_{x_j} h, \dot{x}_j\rangle$ (an inner product of d_x-dimensional vectors); sup means some control achieves the inequality. $\mathcal{B}_i$ is the set of local states where i and its neighbours are currently safe.

Remark 1. Assumption 2 amounts to $\nabla_{x_j} h = 0$ for $j \in \mathcal{N}_i^{\hat{R}} \setminus \mathcal{N}_i^R$ and $h(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}}) = h(x^{\mathcal{N}_i^{R}\cup\{i\}})$.

What does Remark 1 mean?

It does *not* say $h = 0$ beyond R; it says h does not *depend* on agents beyond R. A hard cutoff would make h jump, so each neighbour's influence must fade smoothly (in value and gradient) as its distance approaches R. Example: $h = \text{softmin}_j \min(\|p_i - p_j\|, R) - r$; a neighbour at distance $\geq R$ contributes the constant R, as if absent.

Remark 2. $x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}}$ has discontinuities from topology changes of $\hat{\mathcal{G}}_t^i$, yet a valid distributed CBF is C^1 [15].

Differentiability is asserted, not derived. Since h_ϕ is a learned GNN, smoothness and the zero-gradient property of Remark 1 are requirements that architecture and training should satisfy; the paper does not show how.

3.2 Understanding $\mathcal{B}_i$ and $\mathcal{C}_N$

The $\mathcal{B}_i$ live in spaces of different dimension $|\mathcal{N}_i|+1$ and cannot be intersected directly. $\mathcal{C}_N$ lifts each local condition to $\mathcal{X}^N$: the condition after the bar constrains only the sub-vector $x^{\mathcal{N}_i\cup\{i\}}$ and leaves other components free (equivalently, $\text{proj}_{\mathcal{N}_i\cup\{i\}}(x^{I_N}) \in \mathcal{B}_i$). As $\mathcal{N}_i$ depends on the state, the projection can change as agents move: this is the topology change of Remark 2.

Three robots on a line

$x_1, x_2, x_3 \in \mathbb{R}$, $\mathcal{N}_1 = \{2\}$, $\mathcal{N}_2 = \{1,3\}$, $\mathcal{N}_3 = \{2\}$, $h = \min_{j\in\mathcal{N}_i} |x_i - x_j| - r$. Lifted to $\mathbb{R}^3$, the local conditions are $|x_1 - x_2| \geq r$; both $|x_1 - x_2| \geq r$ and $|x_2 - x_3| \geq r$; and $|x_2 - x_3| \geq r$. Their intersection is $\mathcal{C}_N = \{|x_1 - x_2| \geq r,\ |x_2 - x_3| \geq r\}$. Robots 1 and 3 are not neighbours, so $|x_1 - x_3|$ is unchecked.

The chain $\mathcal{C}_N \subset \bigcap_i S_{N,i}^r \subseteq S_N^r$. The first inclusion holds because $h \geq 0$ means every agent is locally collision-free; this is what makes h a certificate and is enforced by training (Section 4.4). It is *strict on purpose*: $\mathcal{C}_N$ is the set made forward invariant, while S_N^r is only the "no collision" specification and is generally not invariant, since some of its states are already doomed.

A doomed but collision-free state

Two robots on a line with $|a| \leq 1$, gap $d > r$, closing speed v. Even with full braking the gap shrinks by $v^2/4$; if $d - r < v^2/4$ a collision is unavoidable. The state is in S_N^r but cannot be in $\mathcal{C}_N$, since Eq. (2) would demand a non-existent control.

Because the invariant set is intractable to compute, the paper labels data with a finite-horizon approximation (Section 4.3), and argues that the more of S_N that $\mathcal{C}_N$ covers, the less restrictive the controller (Section 5.5).

3.3 Lemma 1: forward invariance and convergence

Lemma 1. Assume a valid CBF h with $\mathcal{C}_N \subseteq D \subset S_N^r$ and a locally Lipschitz policy $\pi : \mathcal{X}^N \to \mathcal{A}^{\text{CBF}}$. For any N, if $x^{I_N}(0) \in \mathcal{C}_N$ then $x^{I_N}(t) \in \mathcal{C}_N \subset S_N^r$ for all t [15]. If $\mathcal{C}_N$ is compact, it is forward invariant and $x^{I_N}(t) \to \mu \in \mathcal{C}_N$ asymptotically if $x^{I_N}(0) \in \overline{\mathcal{C}}_N \cap D$ [7], [12], [25].

Reading the lemma. D is a region containing $\mathcal{C}_N$ on which h is defined. $\mathcal{A}^{\text{CBF}}$ is the set of controls satisfying Eq. (2); local Lipschitz continuity gives existence and uniqueness of trajectories. $\overline{\mathcal{C}}_N$ is undefined in the paper; I read it as the *complement* of $\mathcal{C}_N$ (a closure would be redundant, as $\{h \geq 0\}$ is already closed), so $\overline{\mathcal{C}}_N \cap D$ is "just outside $\mathcal{C}_N$ but still collision-free". Compactness adds boundedness, which prevents drift to infinity and yields a limit point; without it safety still holds but convergence may fail.

Nagumo vs. Comparison Lemma

The paper notes $h = 0$ on $\partial\mathcal{C}_{N,i}$ is *not required* if the Comparison Lemma replaces Nagumo's theorem [24] ($\mathcal{C}_{N,i}$ is undefined; I read it as $\mathcal{B}_i$). Nagumo-style proofs check the vector field at boundary points and need the boundary to equal $\{h = 0\}$. The Comparison Lemma uses only $\dot{h} \geq -\alpha(h)$ along trajectories, which suits a learned h whose zero level set cannot be forced to a prescribed boundary.

4 Learning a Safe Distributed Policy

4.1 Per-agent constrained problem

Applying Lemma 1 replaces $x^{I_N} \in S_N^r$ by the local CBF condition:

$$\min_{u_i(0:T)} \int_0^T \sum_i \mathcal{T}(x_i(\tau))\mathrm{d}\tau \tag{3}$$

$$\text{s.t.} \quad \dot{x}_i = f(x_i, u_i) \in \mathfrak{X}(\mathcal{X}), \quad u_i = \pi\big(x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}}\big) \in \mathcal{U}, \tag{4,5}$$

$$\sum_{j \in \mathcal{N}_i^{\hat{R}} \cup \{i\}} \big\langle \nabla_{x_j} h(x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}}), f(x_j, u_j) \big\rangle \geq -\alpha\big(h(x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}})\big), \quad \forall i. \tag{6}$$

Eq. (3) is the liveness objective and Eq. (6) the safety constraint. The paper writes $\pi(x^{\mathcal{N}_i^{\hat{R}} \cup \{i\}})$ for $\pi(\hat{\mathcal{G}}_t^i)$: a notational identification (the local state determines the graph's agent nodes), imprecise since target and LiDAR nodes are dropped.

Why training is centralised. Eq. (6) involves neighbours' controls u_j, so one agent's safety needs its neighbours' cooperation, giving a non-convex centralised problem that is intractable online. The paper therefore trains on-policy, in *centralised-training, distributed-inference* fashion, a candidate CBF h_ϕ and policy π_θ, both GNNs (suited to graph signals, variable neighbourhood size, permutation equivariant [26]).

On-policy loop. (1) Run the current π_θ in random episodes; (2) sample local graphs and label them safe/unsafe (Section 4.3); (3) solve QP (7) with h_ϕ, π_{nom} to get π_{QP}; (4) compute loss (8), update θ, ϕ. Using the current policy limits distributional shift.

Shared experience. All robots share the same weights and treat agents by relative geometry, not ID (permutation equivariance), so an N-robot episode yields about N useful samples for one shared function. (In training, constraint (6) also couples agent i to neighbours' policies via backpropagation.)

4.2 Constrained QP and liveness

Liveness is obtained by *imitating* a nominal controller $u_{i,\text{nom}} = \pi_{\text{nom}}(x_i, \hat{x}_i)$. Since it has no safety context, the reference is the min-norm safe solution

$$\min_{u_i \in \mathcal{U},\ \forall i} \|u_i - u_{i,\text{nom}}\|^\kappa \quad \text{s.t. Eq. (6)}, \tag{7}$$

a QP for control-affine dynamics and convex $\mathcal{U}$, with solution $\pi_{QP} : \mathcal{X}^N \to \mathcal{U}^N$.

4.3 Training data and labels

The dataset is $\mathcal{D}_i = \mathcal{D}_{S,i} \cup \mathcal{D}_{U,i}$ (safe and unsafe samples of agent i), each sample being the local graph over steps $0..k$ together with the stacked controls of i and its neighbours (needed because Eq. (6) contains $f(x_j, u_j)$). Following the finite reachability approximation of [15]:

- **Unsafe:** $\hat{\mathcal{G}}_t^i$ already contains a collision.
- **Safe:** no collision now *and* none when rolling out π_θ for horizon T (control-invariant demonstrations; $T \to \infty$ recovers the intractable invariant set).
- **Unlabelled:** no collision now but one in the rollout.

Distance-only labelling would not do: a collision-free state can be doomed.

What happens to unlabelled samples?

(Not in the paper, which says only that they "remain unlabeled".) They belong to none of the sets the hinge terms sum over, so they train none of them; whether they enter the imitation term is unstated. They are not labelled unsafe because the rollout cannot tell whether the state is doomed or the policy is merely poor; a similar state may get a label later as the policy improves.

4.4 The loss

Following [13],

$$\mathcal{L}_{\theta,\phi} := \sum_{i\in I_N}\Bigg[\eta_c\big\|\pi_\theta(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}}) - \pi_{QP}(x^{I_N})\big\| + \eta_d\sum_{\mathcal{D}_i}\Bigg[\gamma - \sum_j\big\langle\nabla_{x_j}h_\phi, f(x_j,u_j)\big\rangle - \alpha(h_\phi)\Bigg]_+ + \sum_{\mathcal{D}_{S,i}}\Big[\gamma - h_\phi(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}})\Big]_+ + \sum_{\mathcal{D}_{U,i}}\Big[\gamma + h_\phi(x^{\mathcal{N}_i^{\hat{R}}\cup\{i\}})\Big]_+\Bigg], \tag{8}$$

with $[z]_+ = \max(z,0)$. Terms: (a) imitate the QP; (b) penalise violating the CBF condition by more than margin γ; (c) push $h_\phi \geq \gamma$ on safe samples; (d) push $h_\phi \leq -\gamma$ on unsafe samples. The loss trains h_ϕ to agree with labels that come from outside it (distance check plus rollout); forward invariance is not a property of the data but follows from h_ϕ being a valid CBF, which term (b) trains.

Why must γ not be 0, and how is it set?

With $\gamma = 0$, $h_\phi \equiv 0$ gives zero loss on all terms ($\dot{h} = 0$, $\alpha(0) = 0$): the safe term only demands $h \geq 0$, the unsafe term $h \leq 0$, so a constant zero separates nothing and $\{h \geq 0\}$ becomes the whole space, violating $\mathcal{C}_N \subset S_N^r$. With $\gamma > 0$, $h \equiv 0$ costs γ per sample, so zero loss needs a genuine margin. The margin also gives robustness to labelling and derivative-approximation error.

γ is a fixed hyperparameter (the optimiser updates only θ, ϕ). The paper uses $\gamma = 0.02$, $\eta_d = 0.2$, $\alpha(\cdot) = \cdot$, learning rates 10^{-4} (CBF) and 10^{-5} (policy), without saying how 0.02 was chosen. Too small revives the degenerate solution; too large shrinks $\mathcal{C}_N$ and makes the controller conservative. (Not in the paper: start small and tune on held-out classification accuracy and simulated safety rate.)

5 Symmetry-Infused Safe Policies

The paper introduces group-invariant multi-agent CBFs, argues why symmetry helps, and then builds the *Equivariant Graphormer*, an architecture respecting symmetries through group-induced canonicalising actions.

5.1 Equivariant dynamics

Definition 5 (G-equivariant dynamics)**.** Dynamics are G-equivariant if, for actions $\phi : \mathcal{X}\times G \to \mathcal{X}$ and $\psi : \mathcal{U}\times G \to \mathcal{U}$,

$$\mathrm{d}\phi_g\, f(x_i,u_i) = f\big(\phi_g(x_i), \psi_g(u_i)\big), \qquad \forall g \in G,$$

where $\mathrm{d}\phi_g$ is the differential of the diffeomorphism ϕ_g.

$\mathrm{d}\phi_g$ (the Jacobian) maps tangent vectors at x to tangent vectors at $\phi_g(x)$; it is a derivative in the state, not in time (rotation: $\mathrm{d}\phi_g = R$; translation: identity). It acts on the *output* f (a velocity), while ψ_g acts on the *input* u. The condition says: with $x' = \phi_g(x)$, $u' = \psi_g(u)$, the chain rule gives $\dot{x}' = \mathrm{d}\phi_g f(x,u)$, so the transformed motion is itself a legal motion of the system.

Check

$\dot{x} = u$ in the plane, g a 90° rotation, $x = (1,0)$, $u = (0,2)$: $Rf(x,u) = R(0,2) = (-2,0)$ and $f(Rx,Ru) = (-2,0)$. Adding gravity, $\dot{x} = u + (0,-9.8)$, breaks this since gravity does not rotate with the world, which is why the experiments use a subgroup of $SE(3)$.

5.2 Topology invariance

Assumption 3. The graph topology is G-invariant: if $V_{\hat{\mathcal{G}}_t} \mapsto \mathcal{E}$ then $\phi_g(V_{\hat{\mathcal{G}}_t}) \mapsto \mathcal{E}$ for all g.

Here $\mathcal{E}$ is the edge set produced from the nodes by the distance rule. It holds for Euclidean distances and $g \in SE(n)$ (e.g. $\hat{R} = 5$, robots at $(0,0),(3,0),(10,0)$: only edge $1 \leftrightarrow 2$, unchanged under rotation; scaling by 2 would break it).

Why it matters. h reads only $Q = \mathcal{N}_i^{\hat{R}} \cup \{i\}$, fixed by the edges; under a group action every node moves by the same g and the edges do not change, so h reads the same agents on both sides. This lets Lemma 2 compare $h(x^Q)$ with $h(\oplus_{k\in Q}\phi_g(x_k))$ for the same Q.

5.3 Invariant functions and sets

Definition 6 (G-invariant set)**.** Y is G-invariant under action ρ if $\rho_g(y) \in Y$ for all $y \in Y$, $g \in G$.

A function h is **G-invariant** if $h(\phi_g(x)) = h(x)$; for stacked states G acts on every component with the same g.

Invariant function vs. invariant set

A function keeps its *value*; a set keeps only *membership*. Invariant $h \Rightarrow \{h \geq 0\}$ is an invariant set. The converse fails: the unit disc equals $\{h \geq 0\}$ for $h(x) = (1 - \|x\|^2)(2 + \sin x_1)$, which is not rotation-invariant.

5.4 Exploiting symmetries for safe policy learning

Define safety through a specification c: $S^r_{N,i} := \{\cdot \mid c(x_i, x_j) \geq 0,\ \forall j \in E_{\hat{\mathcal{G}}^i_t} \setminus E_*\}$ (Euclidean: $c = \|p_i - p_j\| - r$). The paper states: if c is G-invariant, $c(\phi_g(\alpha), \phi_g(\beta)) = c(\alpha, \beta)$, then S_N is G-invariant; e.g. the Euclidean specification is $SE(n)$-invariant.

Why is this true?

Membership means $c(x_i, x_j) \geq 0$ for every neighbour; after transformation $c(\phi_g x_i, \phi_g x_j) = c(x_i, x_j) \geq 0$. The neighbour list must also be unchanged, which is Assumption 3 (the paper's sentence omits it). For Euclidean distance with $\phi_g(p) = R_0 p + p_0$: $\|R_0(p_i - p_j)\| = \|p_i - p_j\|$. A non-invariant example is horizontal separation $|\alpha_x - \beta_x| - r$ (a vertical pair is unsafe but safe after a 90° rotation); "stay below altitude 10" is not translation-invariant and needs a smaller group.

Definition 7 (Valid G-invariant decentralised CBF)**.** Given G-equivariant dynamics, a C^1 G-invariant h, Assumptions 3 and 2, and $\mathcal{C}_N \subset S^r_N$ with both sets G-invariant, h is a *valid G-invariant decentralised CBF* if Eq. (2) holds for some locally Lipschitz extended class-$\mathcal{K}$ α.

This is a definition, not a result: it is the hypothesis of Lemma 2 and Theorem 2 and the basis of the invariant network h_ϕ in Section 6.

Lemma 2. For a CBF of Definition 7, the constraint (2) is preserved under G: with $Q = \mathcal{N}_i^{\hat{R}} \cup \{i\}$, if it holds at (x^Q, u^Q) then

$$\sum_{j \in Q} \left\langle \nabla_{\phi_g(x_j)} h\big(\oplus_{k \in Q} \phi_g(x_k)\big), f\big(\phi_g(x_j), \psi_g(u_j)\big) \right\rangle \geq -\alpha\Big(h\big(\oplus_{k \in Q} \phi_g(x_k)\big)\Big), \quad \forall g \in G. \tag{9}$$

Hence Lemma 1 still applies and forward invariance in $\mathcal{C}_N$ is guaranteed.

Lemma 3. For G-equivariant dynamics and a valid non-invariant CBF h on the G-invariant set $C = \{h \geq 0\}$, there exists an equivalent G-invariant valid CBF $\hat{h}$.

Meaning and caveat (not in the paper)

A physical safety requirement (e.g. drones ≥ 0.5 m apart) is a G-invariant set, but the network learns only invariant h_ϕ. Lemma 3 says that, *if* some valid CBF has superlevel set C, an invariant one exists too, so restricting to invariant functions costs no certifying power. It assumes a valid CBF exists. **Caveat:** averaging with a normalised Haar measure needs a compact group, whereas $\bar{G} \cong SE(2) \times \mathbb{R}$ of the experiments is non-compact. $SO(n)$ and $O(n)$ are compact while $\mathbb{R}^n$ and $SE(n)$ are not.

5.5 Why learn G-invariant CBFs?

The paper's argument. Take S_N G-invariant, dynamics equivariant, an invariant valid CBF h and a non-symmetric valid CBF $\hat{h}$, with $\mathcal{C}_N = \{h \geq 0\}$, $\hat{\mathcal{C}}_N = \{\hat{h} \geq 0\}$, both inside S_N. Since $\hat{h}$ lacks symmetry, there are $x_1, x_2 = \phi_g(x_1)$ with $x_1 \in \hat{\mathcal{C}}_N$, $x_2 \notin \hat{\mathcal{C}}_N$: it certifies a situation but not its rotated twin. For invariant h this cannot happen. The paper concludes that *if* $\mathcal{C}_N \supset \hat{\mathcal{C}}_N$, constraint (6) is less restrictive for the invariant CBF, so greater coverage of S_N means better performance.

Why a larger certified set helps

$\dot{h} \geq -\alpha(h)$ restricts the controller most near $h \approx 0$. In a smaller non-symmetric set the robot hits the artificial boundary early and the safety filter intervenes without real risk (e.g. two drones 2 m apart judged safe facing North but nearly unsafe facing East). An invariant CBF treats all rotated and shifted copies identically, reducing deadlocks.

The inclusion is a heuristic, not a theorem

The paper states $\mathcal{C}_N \supset \hat{\mathcal{C}}_N$ conditionally and without proof; it can fail for arbitrary valid CBFs. Precisely (not in the paper): (i) symmetrising by *intersection*, $\bigcap_g \phi_g(\hat{\mathcal{C}}_N)$, loses coverage; (ii) symmetrising by *union*, $\bigcup_g \phi_g(\hat{\mathcal{C}}_N)$, is invariant, contains $\hat{\mathcal{C}}_N$, lies in S_N and is control-invariant, but equals $\{\max_g \hat{h}(\phi_g x) \geq 0\}$, which is not smooth and so need not

admit a smooth valid h; (iii) the largest control-invariant subset $K \subset S_N$ is G-invariant and contains every valid certified set, so an invariant certificate can in principle reach all of K, though a learned h is not guaranteed to.

Smaller hypothesis class. Restricting to invariant functions shrinks the space of learnable functions: fewer parameters, faster convergence and better sample efficiency, since many (x, u) pairs related by the induced actions are safety-equivalent (Fig. 1 of the paper).

Generalisation. One sample teaches its whole orbit: for invariant S_N and equivariant dynamics, an invariant CBF trained on $D = \{(x^k, u^k)\}$ generalises to

$$\hat{D} = \bigcup_{g \in G} \{(\phi_g(x^k), \psi_g(u^k))\}_k, \qquad D \subseteq \hat{D} \subseteq \mathcal{X}^N \ (g = e).$$

This matters in multi-agent systems, where $\mathcal{X}^N$ grows with N while data is limited; quotienting out symmetries shrinks the space to cover. The paper shows zero-shot transfer to swarms of up to 512 robots after training on 8.

5.6 Policy equivariance

The nominal controller solves the obstacle-free problem (a formal way to write a start-to-finish trajectory):

$$u^*_{i,\text{nom}} = \arg \min_{u^i_{0:T} \in \mathcal{U}} \int_0^T \mathcal{T}\big(x_i(\tau) \mid \hat{x}_i\big) \mathrm{d}\tau \quad \text{s.t. } \dot{x}_i = f(x_i, u_i). \tag{10}$$

Theorem 1. For a group G with transitive actions, G-equivariant dynamics and G-invariant cost $\mathcal{T}$, the optimal nominal policy $\pi^*(x_i)$ for (10) is G-equivariant.

Theorem 2. For G-equivariant dynamics, G-invariant cost $\mathcal{T}$ and a valid G-invariant CBF (Definition 7), the optimal policy π_{QP} of (7) is G-equivariant.

Outcome

Both u_{nom} and u_{QP} are equivariant, so π_θ imitates an equivariant target. The paper adds that an equivariant network for π_θ improves generalisation and sample efficiency [27]; this is cited, not proved.

6 The Equivariant Graph Transformer

Goal. Learn a G-equivariant policy $\pi_\theta(\hat{\mathcal{G}}^i_t)$ and a G-invariant CBF $h_\phi(\hat{\mathcal{G}}^i_t)$. Specialised equivariant architectures are hard to optimise, slow and hard to adapt to new manifolds. The authors instead obtain equivariance *via group actions on the graph*, with no extra constraints, extending existing architectures (e.g. [28], [29]).

6.1 Group canonicalisation

Idea. Make any off-the-shelf network (MLP, graph transformer) equivariant by (1) **canonicalising** the input into a fixed reference frame (un-rotate, un-shift), (2) **processing** it with an unconstrained network h, (3) **de-canonicalising** the output back to the original frame.

The extended action on $G \times \mathcal{X}$ is $\phi'_g[(p, x)] = (g \cdot p, \phi_g[x])$, with $p \in G$ a reference frame and x the state; it is again an action.

Lemma 4. $f : G \times \mathcal{X} \to \mathcal{Y}$ satisfies $f(\phi'_g[p]) = \psi_g[f(p)]$ for all g, p, iff $f(g, x) = \psi_g\big[h(\phi_{g^{-1}}[x])\big]$ with $h(x) = f(e, x)$ for some $h : \mathcal{X} \to \mathcal{Y}$.

Reading. g is the agent's local pose. $\phi_{g^{-1}}$ maps the state into the canonical frame (position at the origin, yaw zero; vectors rotated into the frame; scalars unchanged). h sees only canonicalised inputs, so it **need not be equivariant**: an MLP or standard attention layer suffices. ψ_g maps the output back.

2D example

$f(g, x) = R_\theta \,\mathrm{MLP}(R_{-\theta} x)$. Rotating the world by R sends $g \to Rg$, $x \to Rx$, so $f \to R\, R_\theta \,\mathrm{MLP}(R_{-\theta} R^{-1} Rx) = Rf(g, x)$: the output rotates with the input although the MLP knows nothing about rotation.

6.2 The equivariant graph transformer

Tensorial features. Each node carries a frame and a feature, $F_{\text{tens}} = \{(g_v, f_v) \in G \times \mathcal{X}\}$; g_v is its local pose and f_v its state feature (velocity, distance, target encoding). Re-expressing in frame g_n: $\phi'_{g_n g_v^{-1}}[(g_v, f_v)] = (g_n, \phi_{g_n g_v^{-1}}[f_v])$.

Standard local-attention layer.

$$f_v^{(l+1)} = M_v(F^{(l)}) = \mu\big(f_v^{(l)} + A(F^{(l)})_v\big),$$
$$A(F^{(l)})_v = \sum_{p\in\mathcal{N}_v} \mathrm{softmax}\big(f_v^{(l)T} W_Q^T W_K\, f_p^{(l)}\big) W_V\, f_p^{(l)},$$

with μ a feed-forward network and W_Q, W_K, W_V learned matrices; summing over neighbours makes it permutation equivariant.

Equivariant update (Lemma 4 with $h = M_v$).

$$\boxed{f_v^{(l+1)} \leftarrow M_v^{\mathrm{eq}}(g_v, F^{(l)}) = \phi_{g_v}^{(l+1)}\Big[M_v\Big(\phi_{g_v^{-1}}^{(l)}[F^{(l)}]\Big)\Big]}$$

Inner: transform all neighbour features into v's canonical frame. **Middle:** the unchanged standard layer. **Outer:** transform the output back with g_v. The network sees only relative geometry and is blind to absolute pose. Positions use the pose action, velocities only the rotation part, scalars the identity.

Notational gap: where does F_{tens} enter?

The update takes g_v and $F^{(l)}$ separately, and g_v has no layer index. The most consistent reading applies the extended action to the whole set: $\phi'_{g_v^{-1}}[(g_p, f_p)] = (g_v^{-1} g_p, \phi_{g_v^{-1}}[f_p])$, so v's frame becomes e and each neighbour's frame becomes the relative pose $g_v^{-1} g_p$. An alternative re-expresses each feature via $\phi_{g_v g_p^{-1}}[f_p]$; the two agree for world-frame features. The paper does not say which it uses.

Properties. The composed network is edge-preserving (edges E unchanged), isomorphic (structure preserved), permutation-equivariant and end-to-end G-equivariant. The experimental networks are

$$\pi_\theta(\hat{\mathcal{G}}_t^i) = \psi_{g_i} \circ m_{\theta_2} \circ \phi_{g_i^{-1}} \circ M_{\theta_1}(\hat{\mathcal{G}}_t^i), \qquad h_\phi(\hat{\mathcal{G}}_t^i) = m_{\phi_2} \circ \phi_{g_i^{-1}} \circ M_{\phi_1}(\hat{\mathcal{G}}_t^i),$$

with M_{θ_1}, M_{ϕ_1} $\bar{G}$-equivariant graphormers and m_{θ_2}, m_{ϕ_2} standard MLPs. The policy output is mapped back by ψ_{g_i} (equivariant); the barrier has no such map, so it is an invariant scalar.

7 Experimental Details

Environment. N quadrotors:

$$\ddot{p}_i = g + \tfrac{1}{m} R_i F_i, \qquad \dot{\omega}_i = J^{-1}\big(\tau_i - \omega_i \times J\omega_i\big), \qquad \dot{R}_i = \omega_{i,\times} R_i,$$

with mass m, gravity $g = [0, 0, -9.81]^T$, $R_i \in SO(3)$, thrust $F_i = F_3 e_3$, inertia J, torque τ_i. This is control-affine, $\dot{x}_i = f_0(x_i) + f_1(x_i)u_i$, with $x_i = [\rho_i, \xi_i] \in SE(3) \times \mathbb{R}^6$ and $u_i = [\tau_i, F_3]$.

Symmetry group. Gravity breaks full $SE(3)$ symmetry (the paper's phrase "gravity is $O(3)$-invariant" looks like a typo). The system is equivariant under

$$\bar{G} = \left\{ \begin{bmatrix} R_z(\theta) & \lambda \\ 0 & 1 \end{bmatrix} \;\middle|\; (\lambda, \theta) \in \mathbb{R}^3 \times S^1 \right\} \cong SE(2) \times \mathbb{R}, \tag{11}$$

i.e. arbitrary 3D translation λ plus yaw θ, which leave pitch and roll relative to gravity unchanged. $\bar{G}$ acts by $\phi_g(x) = (g\rho, \xi)$ on states and $\psi_g(u) = (g\tau, F_3)$ on inputs: torque has a direction and is transformed, thrust magnitude is a scalar. (As printed, the twist is left unchanged, so torque and thrust are read as local-frame quantities; this must be read together with the rotation of $\dot{p}$ in Section 2.3 when $\dot{p}$ is in the world frame. The paper does not elaborate.)

With Euclidean neighbourhoods, Assumption 3 holds for $\bar{G}$ since shifts and yaw rotations preserve all $\|p_i - p_j\|$. Each node's pose gives a group element $g_v \in \bar{G}$, so features are $(g_v, f_v) \in \bar{G} \times \mathcal{X}$.

Data and training. Initial, target and obstacle positions are uniform over $[0, l]^3$. Data is collected on-policy with the finite reachability labelling of Section 4.3. A $\bar{G}$-equivariant π_θ and a $\bar{G}$-invariant h_ϕ are trained in parallel with Eq. (8), with π_{QP} from the centralised QP-CBF. Two variants: EGCBF+ and EGCBF, using π_{QP} and π_{nom} respectively as the reference in the loss. Baselines: GCBF+ [15], GCBF [14] without policy refinement, and centralised/decentralised CBF-QPs. Adam, $l = 2$, learning rates 10^{-4} (CBF) and 10^{-5} (policy), $\gamma = 0.02$, $\eta_d = 0.2$, $\alpha(\cdot) = \cdot$; other hyperparameters follow [15].

Scalability. Trained with 8 robots, $l = 2$, 5 obstacles; tested zero-shot on 8 to 512 robots with $l = 4$. Each quadrotor must reach its target while avoiding obstacles and others. Metrics: safety rate, reach rate, success rate, collisions per episode per agent, cumulative reward.

Summary

The paper in six lines

1) The swarm is a graph; poses are group elements (a torsor), everything else transforms compatibly with the group.
2) A shared barrier h certifies each agent's neighbourhood; the lifted intersection $\mathcal{C}_N$ lies inside the collision-free set and is made forward invariant by $\dot{h} \geq -\alpha(h)$ (Lemma 1).
3) Training is on-policy and centralised: h_ϕ, π_θ are learned jointly with the hinge loss (8), imitating the QP (7); execution is distributed.
4) With equivariant dynamics and symmetric topology (Assumption 3), an invariant h keeps the CBF constraint valid under group actions (Lemma 2) and the optimal policy is equivariant (Theorems 1, 2).
5) Symmetric certificates cover more of the safe set, need fewer parameters, and generalise from one sample to its whole orbit.
6) Equivariance comes from canonicalisation (Lemma 4): express features in each node's frame, apply a standard graph transformer, transform back.